d'Alembert's Formula and Finite Propagation

An End-to-End Meta-Math Walkthrough

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Abstract

Characteristic coordinates factor the wave operator. The initial displacement fixes the sum of the two traveling waves; the initial velocity fixes their derivative difference. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PDE-05 / Partial Differential Equations / Report v1

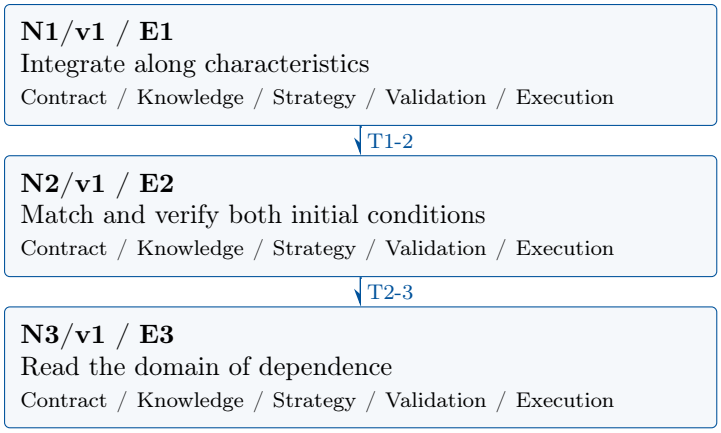
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For $c > 0$, $f \in C^2(\mathbb{R})$ and $g \in C^1(\mathbb{R})$, prove that the unique C^2 solution on $\mathbb{R} \times [0, \infty)$ of $u_{tt} = c^2 u_{xx}$, $u(x, 0) = f(x)$, $u_t(x, 0) = g(x)$, is $u(x, t) = \frac{1}{2}[f(x - ct) + f(x + ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)ds$. Prove that $u(x, t)$ depends only on the initial data in $[x - ct, x + ct]$; compactly supported initial data in $[-R, R]$ yield support inside $[-R - ct, R + ct]$.*

A. Foundations and permitted imports

Use the chain rule, the fundamental theorem for variable-endpoint integrals, and integration of $U_{\xi\eta} = 0$ in characteristic coordinates. The latter integration is proved below on the half-plane corresponding to $t > 0$; no assumption that the solution separates globally is imported. C^2 at $t = 0$ means the derivatives extend continuously from the half-plane.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. Every candidate has the form $u(x,t) = A(x-ct) + B(x+ct)$ for $t > 0$, with C^2 functions A, B.

Receiving activity. N2.1

Compatibility. E1/v1 and f, g on the initial line. Check: The initial velocity fixes opposite signs for the two waves; the integral coefficient is $1/(2c)$.

T2-3: N2/E2/v1 to N3. The displayed formula is the only possible solution and is a C^2 solution with the prescribed data.

Receiving activity. N3.1

Compatibility. E2/v1 and the explicit endpoints and integral interval. Check: Finite propagation is a support statement in space-time; it does not assert that the solution itself stays constant inside the cone.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Integrate along characteristics

Object and role. Factor the differential operator and integrate its vanishing mixed derivative on $\eta > \xi$. **Local obligation:** Every candidate has the form $u(x,t) = A(x-ct) + B(x+ct)$ for $t > 0$, with C^2 functions A, B.

Inputs. Any C^2 candidate solution, with $\xi = x - ct$, $\eta = x + ct$, and $U(\xi,\eta) = u((\xi+\eta)/2, (\eta - \xi)/(2c))$.

C — Contract

Publish E1/v1 only after establishing this obligation: Every candidate has the form $u(x,t) = A(x-ct) + B(x+ct)$ for $t > 0$, with C^2 functions A, B.

K — Knowledge

Any C^2 candidate solution, with $\xi = x - ct$, $\eta = x + ct$, and $U(\xi,\eta) = u((\xi+\eta)/2, (\eta - \xi)/(2c))$. Retain the stated assumptions and import versions.

S — Strategy

Factor the differential operator and integrate its vanishing mixed derivative on $\eta > \xi$.

V — Validation

The transformed domain is a half-plane; justify separation there rather than presuming a rectangular global domain.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 The chain rule gives $u_{tt} - c^2 u_{xx} = -4c^2 U_{\xi\eta}$, so the mixed derivative vanishes on $\eta > \xi$. For fixed ξ, $U_\xi(\xi,\eta)$ is constant on the connected interval $\eta > \xi$. Denote it by $a(\xi)$; locally one can evaluate at a fixed admissible η, so a is C^1.

N1.2 Choose an antiderivative A of a. Then $U - A$ has zero ξ derivative. For each fixed η, the interval $\xi < \eta$ is connected, so $U(\xi,\eta) - A(\xi)$ depends only on η; call it $B(\eta)$. Local evaluation shows B is C^2. Thus $U = A + B$ on the half-plane. Continuity extends the corresponding identities to $t = 0$.

Export E1/v1. Every candidate has the form $u(x,t) = A(x-ct) + B(x+ct)$ for $t > 0$, with C^2 functions A, B.

Acceptance gate. The transformed domain is a half-plane; justify separation there rather than presuming a rectangular global domain.

Targeted repair. If separation is asserted without checking the transformed domain, use connected characteristic intervals to integrate each derivative. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Match and verify both initial conditions

Object and role. Solve for the derivatives of the traveling-wave components and substitute their antiderivatives. **Local obligation:** The displayed formula is the only possible solution and is a C^2 solution with the prescribed data.

Inputs. E1/v1 and f, g on the initial line. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: The displayed formula is the only possible solution and is a C^2 solution with the prescribed data.

K — Knowledge

E1/v1 and f, g on the initial line. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Solve for the derivatives of the traveling-wave components and substitute their antiderivatives.

V — Validation

The initial velocity fixes opposite signs for the two waves; the integral coefficient is $1/(2c)$.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 At $t = 0$, $A(x) + B(x) = f(x)$ and $-cA'(x) + cB'(x) = g(x)$. Differentiating the first identity and solving gives $A' = (f' - g/c)/2$, $B' = (f' + g/c)/2$. If $G(x) = \int_0^x g(s)ds$, integration gives $A = f/2 - G/(2c) + C$ and $B = f/2 + G/(2c) - C$.

N2.2 Substitution at $x - ct$ and $x + ct$ cancels C and gives d'Alembert's formula. Conversely these A, B are C^2 because $f \in C^2$ and $G \in C^2$. Each traveling wave satisfies $u_{tt} = c^2 u_{xx}$ by direct differentiation, and the two initial identities hold. Thus a solution exists and every candidate equals it, proving uniqueness.

Export E2/v1. The displayed formula is the only possible solution and is a C^2 solution with the prescribed data.

Acceptance gate. The initial velocity fixes opposite signs for the two waves; the integral coefficient is $1/(2c)$.

Targeted repair. If only the initial displacement is matched, compute the time derivative at zero and fix the relative sign and factor using g. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Read the domain of dependence

Object and role. Locate every initial value used by the solution and compare two data sets on that interval. **Local obligation:** Only initial data in $[x - ct, x + ct]$ affect $u(x, t)$, and the stated support bound holds.

Inputs. E2/v1 and the explicit endpoints and integral interval. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: Only initial data in $[x - ct, x + ct]$ affect $u(x, t)$, and the stated support bound holds.

K — Knowledge

E2/v1 and the explicit endpoints and integral interval. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Locate every initial value used by the solution and compare two data sets on that interval.

V — Validation

Finite propagation is a support statement in space-time; it does not assert that the solution itself stays constant inside the cone.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 The formula uses f only at the two endpoints and g only inside the closed interval. Therefore two admissible initial data sets agreeing on that interval produce the same value at (x, t).

N3.2 If f, g vanish outside $[-R, R]$ and $x > R + ct$, then the entire integration interval and both endpoints lie to the right of R, so every term is zero. The same holds for $x < -R - ct$. Hence the support at time t is contained in the enlarged interval, proving finite propagation.

Export E3/v1. Only initial data in $[x - ct, x + ct]$ affect $u(x, t)$, and the stated support bound holds.

Acceptance gate. Finite propagation is a support statement in space-time; it does not assert that the solution itself stays constant inside the cone.

Targeted repair. If the support is claimed to stay inside its initial interval, expand the interval by the characteristic distance ct. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Integrate along characteristics

1. **N1.1.** The chain rule gives $u_{tt} - c^2 u_{xx} = -4c^2 U_{\xi\eta}$, so the mixed derivative vanishes on $\eta > \xi$. For fixed ξ, $U_\xi(\xi, \eta)$ is constant on the connected interval $\eta > \xi$. Denote it by $a(\xi)$; locally one can evaluate at a fixed admissible η, so a is C^1.

2. **N1.2.** Choose an antiderivative A of a. Then $U - A$ has zero ξ derivative. For each fixed η, the interval $\xi < \eta$ is connected, so $U(\xi, \eta) - A(\xi)$ depends only on η; call it $B(\eta)$. Local evaluation shows B is C^2. Thus $U = A + B$ on the half-plane. Continuity extends the corresponding identities to $t = 0$.

B. N2 — Match and verify both initial conditions

1. **N2.1.** At $t = 0$, $A(x) + B(x) = f(x)$ and $-cA'(x) + cB'(x) = g(x)$. Differentiating the first identity and solving gives $A' = (f' - g/c)/2$, $B' = (f' + g/c)/2$. If $G(x) = \int_0^x g(s)ds$, integration gives $A = f/2 - G/(2c) + C$ and $B = f/2 + G/(2c) - C$.

2. **N2.2.** Substitution at $x - ct$ and $x + ct$ cancels C and gives d'Alembert's formula. Conversely these A, B are C^2 because $f \in C^2$ and $G \in C^2$. Each traveling wave satisfies $u_{tt} = c^2 u_{xx}$ by direct differentiation, and the two initial identities hold. Thus a solution exists and every candidate equals it, proving uniqueness.

C. N3 — Read the domain of dependence

1. **N3.1.** The formula uses f only at the two endpoints and g only inside the closed interval. Therefore two admissible initial data sets agreeing on that interval produce the same value at (x, t).

2. **N3.2.** If f, g vanish outside $[-R, R]$ and $x > R + ct$, then the entire integration interval and both endpoints lie to the right of R, so every term is zero. The same holds for $x < -R - ct$. Hence the support at time t is contained in the enlarged interval, proving finite propagation.

□

V. WORKED ORCHESTRATION

For $c = 1$, $f = 0$, and $g(s) = \cos s$, the formula gives $u(x, t) = (\sin(x + t) - \sin(x - t))/2 = \cos x \sin t$. Then $u_{tt} = -\cos x \sin t = u_{xx}$, $u(x, 0) = 0$, and $u_t(x, 0) = \cos x$.

A. Boundary and hypothesis audit

Initial displacement alone does not determine a wave solution. Both $u = 0$ and $u = t$ have zero initial displacement and satisfy the wave equation, but their initial velocities differ. Prescribing both f and g is essential.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 solution: keep only $[f(x - ct) + f(x + ct)]/2$ even when $g \neq 0$. Its time derivative at zero is zero, so it fails the velocity datum. **Diagnosis:** Execution matched only half the initial contract. **Repair:** solve the two derivative equations and add the integral of g with coefficient $1/(2c)$. Recheck the PDE, both initial data, and the interval used for finite propagation.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Extend characteristic reasoning to inhomogeneous forcing with Duhamel's principle, using a new source-term contract.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Leon Simon, Lectures on PDE, for elliptic, weak, and parabolic methods. Classical calculus and the explicitly stated Hilbert-space imports are used below; the displayed PDE arguments are developed in full.